# Marks Average Program

Java Lab Record

## 1. Aim

To develop a Java program that accepts marks of five subjects from the user, calculates the total marks, and finds the average marks using Scanner and arithmetic operations.

## 2. Step-by-Step Procedure

1. Import the Scanner class to accept input from the user.
2. Create the MarksAverage class containing the main() method.
3. Create a Scanner object for reading input.
4. Read the marks of Subject 1.
5. Read the marks of Subject 2.
6. Read the marks of Subject 3.
7. Read the marks of Subject 4.
8. Read the marks of Subject 5.
9. Calculate the total marks by adding the five subject marks.
10. Calculate the average by dividing the total marks by 5.0.
11. Display the total marks.
12. Display the average marks.
13. Close the Scanner object.
14. Stop the program.

## 3. Algorithm

1. Start.
2. Import java.util.Scanner.
3. Create a Scanner object.
4. Read marks of five subjects as m1, m2, m3, m4, and m5.
5. Calculate total = m1 + m2 + m3 + m4 + m5.
6. Calculate average = total / 5.0.
7. Display the total marks.
8. Display the average marks.
9. Close the Scanner object.
10. Stop.

## 4. UML Class Diagram

```
+---------------------------------------+
|          MarksAverage                 |
+---------------------------------------+
|                                       |
| main(String[] args)                   |
+---------------------------------------+
              |
              | uses
              v
+---------------------------------------+
|              Scanner                  |
+---------------------------------------+
| nextInt()                             |
| close()                               |
+---------------------------------------+
```

## 5. Code

```
import java.util.Scanner;

public class MarksAverage {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks of Subject 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter marks of Subject 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter marks of Subject 3: ");
        int m3 = sc.nextInt();

        System.out.print("Enter marks of Subject 4: ");
        int m4 = sc.nextInt();

        System.out.print("Enter marks of Subject 5: ");
        int m5 = sc.nextInt();

        int total = m1 + m2 + m3 + m4 + m5;
        double average = total / 5.0;

        System.out.println("Total Marks = " + total);
        System.out.println("Average Marks = " + average);

        sc.close();
    }
}
```

## 6. Sample Output

Sample execution using marks 80, 75, 90, 85, and 70. The output is formatted in the same console-style presentation as the reference.

```
--- Marks Average Program ---

Enter marks of Subject 1: 80

Enter marks of Subject 2: 75

Enter marks of Subject 3: 90

Enter marks of Subject 4: 85

Enter marks of Subject 5: 70

Total Marks = 400

Average Marks = 80.0
```